

Robust Filtering for Multi-Object Tracking Against Stealthy Measurement-Oriented Adversarial Attacks

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Abstract

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Index Terms—Distributed systems, multi-object tracking, resilient filtering, large-scale networks, labeled multi-Bernoulli, random finite set.

I. INTRODUCTION

Multi-object tracking involves estimating both the number of objects and their trajectories from sensor data, while accounting for variations and noise in the observations [1]–[3]. MOT has a diverse range of applications, including vehicle tracking and control [4], robotics [5], surveillance [6], computer vision [7]–[11] and cell biology [12], [13]. To address the challenges of false alarms, missed detections, and a randomly varying number of targets, various approaches to MOT have been developed [14]. Most of these approaches align with three principal frameworks: joint probabilistic data association (JPDA) [1], [15], multiple hypothesis tracking (MHT) [16], and random finite set (RFS) theory [17], [18].

The RFS framework has gained significant attention for its effectiveness in managing complex tracking scenarios over the last two decades. Given its robust mathematical foundation, several RFS-based filters have emerged, including the Probability Hypothesis Density (PHD) filter [19], Cardinalized PHD (CPHD) filter [20], multi-Bernoulli filter [21], labeled multi-Bernoulli (LMB) filter [22] and the generalised version of LMB filter called Vo-Vo filter (GLMB) [23], [24].

Based on the theory of labeled RFSs, the GLMB filter represents the first exact closed-form solution for multi-object tracking, efficiently approximated using Gibbs sampling [1], [25]. The GLMB is a versatile model that balances tractability and fidelity well. By leveraging the conjugacy of the GLMB family with the standard measurement model, the filter propagates the labeled multi-object filtering density forward in time

[24]. Each iteration of the GLMB filter involves a prediction operation and an update operation, producing weighted sums of multi-object exponentials with an intractably large number of terms.

The initial implementation of the GLMB filter truncates these sums using the K-shortest path and ranked assignment algorithms during the prediction and update stages to identify the most significant components [23]. An efficient implementation of GLMB (δ -GLMB) with linear complexity in the number of measurements is presented in [2], featuring a joint prediction-update process and a Gibbs sampling-based truncation technique for effective data association and ranked assignment, thereby significantly reducing computational costs. A notable alternative of GLMB is the Labeled Multi Bernoulli (LMB) filter [22], which attains efficiency by approximating the GLMB filter at the level of its first order statistical moment.

Due to its reliability and adaptability, the GLMB/LMB filter has been applied across various domains, including lineage tracking [26], [27], track-before-detect [28], [29], distributed MOT [30], [31], path planning [32]–[34], multi-sensor [35], multi-scan [35], and large-scale MOT [36]. To enhance the application of labeled RFS tracking methods, there has been growing interest in employing them within multi-sensor multi-object tracking (MSMOT) systems [37]–[39]. However, existing MSMOT filtering algorithms are generally developed under the assumption that sensor measurements are accurate, accounting only for measurement noise.

Consider a stealthy ghost attack scenario where the attacker injects false measurements corresponding to imaginary objects by exploiting the motion model of real objects or performs replay attacks by replicating the measurements of real objects. When false measurements closely mimic the behavior of real objects, the filter is prone to mistakenly associating them with actual targets. This misassociation disrupts the tracking process, resulting in inaccurate target estimates. Additionally, the filter’s performance suffers because it must process an increased number of measurements, including the false ones, which adds computational overhead and causes delays. Together, these factors degrade the overall accuracy and efficiency of the filter.

Also, ghost incidents (or attacks) can occur unintentionally due to environmental situations, for instance, in scenarios where radar signals reflect off nearby surfaces such as concrete walls, metallic objects, or other vehicles. These reflections create multi-path propagation, causing the radar to receive delayed or altered echoes [40]. As a result, the system might misinterpret these echoes as legitimate detections, leading to the appearance of false or ghost targets. Such scenarios are

common in environments with high reflectivity, like urban areas, tunnels, or roadways flanked by large reflective structures. This unintentional phenomenon can impact the accuracy of perception systems in vehicles, even in the absence of a deliberate attacker.

In this context, this paper introduces an attack-resilient MSMOT filtering algorithm, **ALARM** (Average Likelihood for Attack-Resilient Multi-Object Filtering), which enhances measurement likelihood estimation during the filter update step by incorporating the collective multi-state prior of each sensing node. Additionally, a stealthy ghost attack generation algorithm is presented that exploits the target's motion model. The proposed method is evaluated with different numbers of ghost objects, assessing its impact on tracking accuracy and the overall computational efficiency of the tracking process. To the best of our knowledge, this is the first attempt to develop a robust MSMOT algorithm specifically designed to counter measurement-oriented attacks targeting the filtering process in a multi-sensor environment.

The main contributions of this paper are:

1) **Development of ALARM: A Novel Attack-Resilient Filtering Method**

The ALARM filtering method is designed to enhance robustness in *multi-sensor multi-object tracking*. The method effectively rejects false measurements at an early stage of the *Bayesian update process*, mitigating the impact of stealthy *ghost attacks* while ensuring real-time efficiency.

2) **First Formal Treatment of Measurement-Oriented Adversarial Attacks in MSMOT**

This work proposes the **first** MSMOT filtering algorithm explicitly designed to counter *measurement-oriented adversarial attacks*, particularly *stealthy ghost attacks* that exploit motion models. A novel *ghost attack generation model* is introduced, enabling rigorous evaluation of *attack resilience* in tracking systems under varying attack intensities.

3) **Enhanced Computational and Tracking Performance in High-Density Attack Scenarios**

The proposed ALARM filter achieves significant improvements in both *tracking accuracy* and *computational efficiency* compared to standard filtering methods. The method ensures scalability and resilience in *large-scale sensor networks*, preventing potential denial-of-service (DoS) caused by adversarial measurement injections.

The remainder of the paper is organized as follows: Section II presents an overview of the related methods for MSMOT. Section III sets up the problem by defining the sensor network characteristics, describing the multi-object system and its observation model, and reviewing the standard filtering methodology. Section IV describes the attack model and the ghost generation algorithm. Section V presents the proposed method for attack-resilient multi-object tracking. Section VI discusses the implementation details of the filter. Section VII evaluates the method through simulations and presents the results. Finally, Section VIII concludes the paper and suggests directions for future research.

II. RELATED WORKS

RFS-based Multi-sensor MOT approaches are generally categorized as centralized or distributed. Centralized methods achieve high estimation accuracy by processing all sensor data collectively [35], [41]. However, they face scalability challenges due to the NP-hard nature of the measurement-to-track association problem [41].

In contrast, distributed methods, which delegate tracking tasks to local sensors, have become increasingly preferable in modern systems due to their scalability and ability to handle large volumes of data efficiently. While they may sacrifice some accuracy compared to centralized approaches, distributed methods offer significant advantages in terms of robustness and real-time performance. As a result, there is a growing trend towards using distributed approaches for MSMOT, driven by the need for scalable, real-time solutions in large-scale applications. Nevertheless, distributed RFS-based tracking remains an evolving field, with many practical challenges yet to be addressed before large-scale, real-time systems can be fully deployed [37].

Multi-sensor multi-object tracking as a cyber physical system (CPS) faces vulnerabilities to cyber attacks (e.g., deception, replay, DoS) when edge sensors relay raw observations over low power wireless links [39], [42]. Processing nodes typically interconnect through dedicated Local Area Network (LAN) or Virtual Private Network (VPN) channels secured with authentication and encryption measures. While significant efforts have been made to design attack-resilient (AR) distributed tracking approaches to counteract or mitigate the effects of unintentional or intentional data modifications, most existing AR methods are primarily focused on state estimation problems involving a single object, where each sensor generates only one measurement per time instant [43]–[48]. Although some recent studies [38], [39] have extended these approaches to multi-object scenarios with multiple measurements at each time step, they still fail to address the challenge of measurement-oriented adversarial attacks.

Measurement-oriented attacks against MSMOT systems can range from basic modifications, deletions, and injections to more sophisticated adversarial ghost attacks [39], [49]. Stealthy ghost attacks are particularly dangerous as they mimic the behavior of real targets, thereby easily compromising the filtering process. Furthermore, due to the sharp increase in computational cost with the number of tracks, an attacker can substantially overwhelm processing capacity by injecting a high volume of ghost measurements. This can degrade real-time tracking performance and potentially lead to a denial-of-service (DoS) scenario. However, to the best of our knowledge, this issue remains unaddressed in multi-sensor multi-object tracking environments.

III. BACKGROUND AND PROBLEM STATEMENT

Symbols and notation used in this paper are summarized in Table I.

A. Sensor Network Model

This work addresses the challenge of MSMOT within a sensor network exposed to adversarial attacks. The sensor

network is represented as an undirected graph (V, E) , where V is the set of nodes and E is the set of edges. A link $(i, j) \in E$ indicates bidirectional communication, meaning that nodes i and j can exchange information with each other. For each node $i \in V$, its neighbors, represented as N_i , are the nodes directly connected to i by an edge in E . Each sensor node is equipped with the following capabilities: capturing observations, performing data processing, transmitting messages, and receiving information from its in-neighboring nodes.

The following assumptions are adopted:

- Two neighboring nodes can exchange their multi-object tracking information exclusively in the form of a multi-object density.
- The measurements at every node are vulnerable to adversarial attacks (the attack model is elaborated in Section IV).

B. Motivating Example: Wide-Area Monitoring System

An analogy with Wide Area Monitoring Systems (WAMS) in large-scale infrastructures motivates the distributed multi-sensor multi-object tracking (MSMOT) framework [50], [51]. In a typical WAMS (see Fig. 2 in [42]) remote sensors send raw measurements to regional processing centers over non secure links that may be subject to attack, while communications among those centers use dedicated secure channels. Likewise, in our peer to peer object tracking network, geographically dispersed sensors send observations to local processing nodes through potentially vulnerable links, whereas processing nodes exchange multi object densities directly with one another over trusted secure connections. This arrangement reflects practical constraints in wide area deployments: securing every sensor to node link can be prohibitively expensive or technically infeasible, yet securing peer communications among processing nodes is essential for resilient decentralized state estimation.

C. Multi-Object System

Let the multi-object state be

$$\mathbf{X} = \{(x_1, \ell_1), \dots, (x_N, \ell_N)\},$$

where each $x_n \in \mathbb{X}$ and $\ell_n \in \mathbb{L}$ denote the kinematic state and label of the n -th object, $n = 1, \dots, N$. We model $\mathbf{X}$ as a LMB RFS whose density is [52]

$$\pi(\mathbf{X}) = \prod_{\ell \in \mathbb{L}} (1 - r^{(\ell)}) \times \prod_{n=1}^N \frac{r^{(\ell_n)} p^{(\ell_n)}(x_n)}{1 - r^{(\ell_n)}}. \quad (1)$$

Here $\mathbb{L}$ is the finite label space, $r^{(\ell)} \in [0, 1]$ is the existence probability of the Bernoulli component (BC) with label ℓ , and $p^{(\ell)}(x)$ is its spatial probability density function (SPDF).

Each labeled BC $\pi^{(\ell)}$ induces a Bernoulli RFS distributed as

$$\pi^{(\ell)}(\mathbf{X}) = \begin{cases} 1 - r^{(\ell)}, & \mathbf{X} = \emptyset, \\ r^{(\ell)} p^{(\ell)}(x), & \mathbf{X} = \{(x, \ell)\}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

so that $\pi^{(\ell)}$ is fully specified by its existence probability $r^{(\ell)}$ and SPDF $p^{(\ell)}(x)$.

D. Standard LMB Filtering

Each object is assigned a unique label $\ell = (k, i)$, where k denotes the time at which the object entered the surveillance region, and i is an index distinguishing objects that enter at the same time. The label space $\mathbb{L}_k$ for objects born at time k is then $\{k\} \times \mathbb{N}$, and the label space $\mathbb{L}_{0:k}$ for objects at time k (including those born prior to k) can be constructed recursively by $\mathbb{L}_{0:k} = \mathbb{L}_{0:k-1} \cup \mathbb{L}_k$.

At time k , each object has a state $x \in \mathbb{X}$ and a label $\ell \in \mathbb{L}_{0:k}$. The multi-object state $\mathbf{X}$ is a finite subset of $\mathbb{X} \times \mathbb{L}_{0:k}$. This construct implies that we can distinguish surviving and birth objects from their labels. Without explicit reference to the time index, let $\mathbb{L}$ denote the label space for the current time, and $\mathbb{B}$ the label space for objects born at the next time. Note that $\mathbb{L}$ and $\mathbb{B}$ are disjoint, i.e., $\mathbb{L} \cap \mathbb{B} = \emptyset$.

Remark 1. The LMB filter is a computationally efficient approximation of the δ -GLMB filter (see Appendix A for a detailed description of the δ -GLMB). In practical implementations of the LMB filter, an LMB approximation of the δ -GLMB update, as proposed in [23], [24], is propagated. Therefore, the predicted density in the LMB filter will first be converted to a δ -GLMB representation, followed by a δ -GLMB update step. For details on the LMB **prediction step** and its conversion to δ -GLMB, the reader is referred to [53]. **Update step:** If the multi-object prior is a δ -GLMB of the form (44), then the multi-object posterior is also a δ -GLMB with the following form [24]

$$\pi(\mathbf{X}|Z) = \Delta(\mathbf{X}) \sum_{(I, \xi) \in \mathcal{F}(L) \times \Xi} \sum_{\theta \in \Theta} \omega^{(I, \xi, \theta)}(Z) \times \delta_I(\mathcal{L}(\mathbf{X})) [p^{(\xi, \theta)}(\cdot|Z)]^{\mathbf{X}} \quad (3)$$

where Θ is the space of mappings $\theta : \mathbb{L} \rightarrow \{0, 1, \dots, |Z|\}$, such that $\theta(i) = \theta(i') > 0$ implies $i = i'$, and

$$\omega^{(I, \xi, \theta)}(Z) = \frac{\delta_{\theta^{-1}(\{0:|Z|\})}(I) \omega^{(I, \xi)}[\eta_Z^{(\xi, \theta)}]^I}{\sum_{(I, \xi) \in \mathcal{F}(L)} \sum_{\theta \in \Theta} \delta_{\theta^{-1}(\{0:|Z|\})}(I) \omega^{(I, \xi)}[\eta_Z^{(\xi, \theta)}]^I}, \quad (4)$$

$$p^{(\xi, \theta)}(x, \ell|Z) = \frac{p^{(\xi)}(x, \ell) \psi_Z(x, \ell; \theta)}{\eta_Z^{(\xi, \theta)}(\ell)}, \quad (5)$$

$$\eta_Z^{(\xi, \theta)}(\ell) = \left\langle p^{(\xi)}(\cdot, \ell), \psi_Z(\cdot, \ell; \theta) \right\rangle, \quad (6)$$

$$\psi_Z(x, \ell; \theta) = \delta_0(\theta(\ell)) q_D(x, \ell) + (1 - \delta_0(\theta(\ell))) \frac{p_D(x, \ell) g(z_{\theta(\ell)}|x, \ell)}{\kappa(z_{\theta(\ell)})}. \quad (7)$$

E. Observation model

Detections at time step k are independent and identically distributed (i.i.d.). At time k , sensor i detects the target $\mathbf{x} = (x_k, \ell)$, if present, with probability $p_D^{(i)}(\mathbf{x})$. If a detection occurs, the sensor generates a measurement $z_k^{(i)} \in \mathbb{R}^{n_z^{(i)}}$ according to

$$z_k^{(i)} = h_k^{(i)}(x_k) + v_k^{(i)}, \quad v_k^{(i)} \sim \mathcal{N}(0, R_k^{(i)}), \quad (8)$$

where $h_k^{(i)}: \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_z^{(i)}}$ is the measurement function for sensor i at time k . The vector $x_k \in \mathbb{R}^{n_x}$ denotes the target kinematic state, and the measurement noise $v_k^{(i)} \in \mathbb{R}^{n_z^{(i)}}$ is Gaussian with zero mean and covariance $R_k^{(i)}$. Hence, the target-originated measurement follows the Gaussian kinematic likelihood

$$g_k^{(i)}(z | x_k) = \mathcal{N}(z; h_k^{(i)}(x_k), R_k^{(i)}). \quad (9)$$

The set $W_k^{(i)} \subset Z_k^{(i)}$ of detected points (non-clutter measurements) is a multi-Bernoulli RFS with parameter set $\{(p_D^{(i)}(\mathbf{x}), g_k^{(i)}(\cdot | \mathbf{x})) : \mathbf{x} \in \mathbf{X}_k\}$, i.e., $W_k^{(i)}$ is distributed according to

$$\pi_D(W_k^{(i)} | \mathbf{X}_k) = \{(p_D^{(i)}(\mathbf{x}), g_k^{(i)}(\cdot | \mathbf{x})) : \mathbf{x} \in \mathbf{X}_k\} (W_k^{(i)}). \quad (10)$$

At time k , sensor i collects both target-originated measurements and false alarms (clutter). Let

$$Z_k^{(i)} = W_k^{(i)} \cup Y_k^{(i)}, \quad (11)$$

where $Z_k^{(i)}$ denotes the complete measurement set collected by sensor i at time k , comprising the target-originated detections $W_k^{(i)}$ and the clutter measurements $Y_k^{(i)}$. The set $Y_k^{(i)}$ of false observations, assumed independent of $W_k^{(i)}$, is modelled as a Poisson RFS with intensity function $\kappa(\cdot)$; hence its distribution is given by [24]

$$\pi_K(Y_k^{(i)}) = e^{-\langle \kappa, 1 \rangle} \kappa^{Y_k^{(i)}}. \quad (12)$$

It follows from the FISST (FInite Set STatistics) that the multi-object likelihood is the convolution of π_D and π_K [24], i.e.,

$$g_k^{(i)}(Z_k^{(i)} | \mathbf{X}_k) = \sum_{W_k^{(i)} \subseteq Z_k^{(i)}} \pi_D(W_k^{(i)} | \mathbf{X}_k) \pi_K(Z_k^{(i)} \setminus W_k^{(i)}). \quad (13)$$

Here, $Z_k^{(i)} \setminus W_k^{(i)}$ denotes the set-difference between $Z_k^{(i)}$ and $W_k^{(i)}$.

Remark 2. In distributed MSMOT setting, the filtering step may be followed by a fusion step [54]–[56], where the node combines the filtered multi-object posteriors received from its neighboring nodes with its local multi-object density to achieve a consensus-based fused density. In these cases, local densities are assumed unaltered, while the attacker compromises exchanged densities via the communication channel. Thus, a node's local density is commonly used as a reference. However, this fails when sensors face measurement-oriented attacks. Without a reliable reference density, fusion becomes infeasible, exposing neighboring nodes to attack-induced distortions.

The primary goal of this work is to develop a robust and efficient filtering method that utilizes neighboring multi-object state information to mitigate the impact of adversarial measurement-oriented attacks during the filtering process, thereby enhancing the reliability of the nodes' local filtered densities.

TABLE I
SYMBOLS AND NOTATION

Notation	Description
p_D	Detection probability.
p_S	Survival probability.
x	The kinematic state vector.
$\mathbf{x}$	The kinematic state vector associated with each labeled target.
$\mathbf{X}$	$\mathbf{X} = \{(x_i, \ell_i)\}_{i=1}^n$: Multi-object state, where x_i is the i -th target's state and ℓ_i its label.
$\Delta(\mathbf{X})$	The distinct-label indicator.
$\mathcal{L}(\mathbf{X})$	The set of labels of $\mathbf{X}$.
$z^{(i)}$	Kinematic measurement of the target at node i .
$Z^{(i)}$	Measurement set collected by node i .
n_x	Dimension of the kinematic state vector x .
n_z	Dimension of the measurement vector.
h^X	$h^X \triangleq \prod_{x \in X} h(x)$ (with $h^\emptyset = 1$): exponential notation for multi-object functions.
$\delta_Y(X)$	$\delta_Y(X) \triangleq \begin{cases} 1, & X = Y, \\ 0, & \text{otherwise,} \end{cases}$ Generalized Kronecker delta.
$1_Y(X)$	$1_Y(X) \triangleq \begin{cases} 1, & X \subseteq Y, \\ 0, & \text{otherwise,} \end{cases}$ Inclusion function (for $X = \{x\}$, one writes $1_Y(x)$).
$\langle f, g \rangle$	$\langle f, g \rangle \triangleq \int f(x) g(x) dx$: standard inner product.
$\mathcal{N}(\cdot; \mu, \Sigma)$	Gaussian PDF with mean μ and covariance Σ .

IV. ATTACK MODEL

This work investigates stealthy, measurement-oriented attacks on multi-sensor, multi-object tracking systems. Ghost measurements are generated and injected into targeted sensors to evaluate their impact on tracking performance. In what follows, the observation model under such attacks, and the ghost-generation algorithm are explained.

A. Observation model under ghost attacks

Under a stealthy ghost attack, at each time step k the measurement set of sensor i becomes corrupted by adversarially injected ghost measurements that closely mimic the true objects' behavior. Accordingly, the attacked measurement set may be written as

$$Z_k^{(i)} = \tilde{W}_k^{(i)} \cup Y_k^{(i)}, \quad (14)$$

where

$$\tilde{W}_k^{(i)} = W_k^{(i)} \cup Z_k^{\text{ghost}}$$

is the attacked detection RFS, $Y_k^{(i)}$ denotes the clutter, and Z_k^{ghost} represents the set of adversarial measurements. Assuming the probability of a target being a ghost equals the detection probability, and that true and ghost targets share the same likelihood density, the multi-object likelihood under attack becomes:

$$g_k^{(i)}(Z_k^{(i)} | \tilde{\mathbf{X}}_k) = \sum_{\tilde{W}_k^{(i)} \subseteq Z_k^{(i)}} \pi_D(\tilde{W}_k^{(i)} | \tilde{\mathbf{X}}_k) \pi_K(Z_k^{(i)} \setminus \tilde{W}_k^{(i)}), \quad (15)$$

where $\tilde{\mathbf{X}}_k$ denotes the multi-object state comprising both true and ghost targets. Note that the prior of the form (44) is conjugate under the likelihood (15) (see Proposition 7 in [24]), so the posterior likewise takes the form (3).

B. Ghost generation

In adversarial stealthy ghost attacks, fake sensor readings are injected alongside true measurements to mislead the multi-object tracker. Algorithm 1 outlines the generation of these ghost measurements.

At each time step k , the ghost measurement set Z_k^{ghost} is reinitialized to $\emptyset$. In a *deception* scenario, each ghost track $n = 1, \dots, N_{\text{ghost}}$ is defined by its birth and death times $(t_{n,\text{birth}}, t_{n,\text{death}})$, its initial state $x_{n,0}$, and its motion parameters (e.g., turn rate $w_{n,\text{turn}}$). Whenever $k \in [t_{n,\text{birth}}, t_{n,\text{death}}]$, the state evolves according to the one-step state transition function Φ :

$$x_{n,k} = \Phi^{k-t_{n,\text{birth}}}(x_{n,0}), \quad (16)$$

where Φ^m denotes the m -fold composition of Φ . At each such k , a noisy ghost measurement is generated by

$$z_k^{(n)} = h_k(x_{n,k}) + v, \quad v \sim \mathcal{N}(0, R), \quad (17)$$

and appended to Z_k^{ghost} .

In a *replay* scenario, once $k \geq t_{\text{rep}}$ the attacker retrieves each recorded measurement z from the buffer at time $k - t_{\text{rep}}$ and injects it directly into Z_k^{ghost} :

$$Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup \{z : z \in Z_{\text{buf}}(k - t_{\text{rep}})\}.$$

In a *delay* scenario, the attacker executes either the deception or replay procedure with a very large number of ghosts, flooding the filter with ghost measurements and thereby inducing computational delays or denial-of-service.

A schematic of these measurement-oriented ghost attacks on sensor node i at time k is shown in Fig. 1. Ghost measurements Z_k^{ghost} enter the sensor-to-node channel, while inter-node exchanges of priors remain secure. The attacker is assumed to possess complete knowledge of the target's motion dynamics and of the sensor's measurement model. This awareness enables the attacker to inject fake measurements that mimic the target's movement, increasing the sophistication and stealthiness of the attack. In this paper, the target's motion model is assumed to follow a coordinated turn, with a constant rate of turn.

V. PROPOSED METHOD

A. ALARM Filtering

This section introduces the Average Likelihood for Attack Resilient Multi-object (ALARM) filtering, a consensus-based filtering method designed for efficient and robust multi-object tracking in multi-sensor environments. The method generalizes single-node filtering and simplifies to standard filtering when the system involves only one sensing node. Figure 2 depicts the overall block diagram of the proposed method. At each time step k , the proposed method operates in two key phases: Communication and Filtering.

Algorithm 1 Ghost Attack Simulation at Time k

```

1: Initialization:
2:    $Z_k^{\text{ghost}} \leftarrow \emptyset$ 
3:   choose scenario  $s \in \{\text{deception, replay, delay}\}$ 
4:   if  $s = \text{deception}$  then
5:     for  $n = 1$  to  $N_{\text{ghost}}$  do
6:       initialize  $(t_{n,\text{birth}}, t_{n,\text{death}}, x_{n,0}, w_{n,\text{turn}})$ 
7:     end for
8:   else if  $s = \text{replay}$  then
9:     initialize replay buffer  $Z_{\text{buf}}$ 
10:    set replay delay  $t_{\text{rep}}$ 
11:   else if  $s = \text{delay}$  then
12:    spawn many ghosts (via deception or replay) to simulate delay
13:   end if
14: Generate Ghost Measurements:
15:   if  $s = \text{deception}$  then
16:     for  $n = 1$  to  $N_{\text{ghost}}$  do
17:       if  $k \in [t_{n,\text{birth}}, t_{n,\text{death}}]$  then
18:          $x_{n,k} = \Phi^{k-t_{n,\text{birth}}}(x_{n,0})$ 
19:         sample  $v \sim \mathcal{N}(0, R)$ 
20:          $z_k^{(n)} = h_k(x_{n,k}) + v$ 
21:          $Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup \{z_k^{(n)}\}$ 
22:       end if
23:     end for
24:   else if  $s = \text{replay}$  then
25:     if  $k \geq t_{\text{rep}}$  then
26:       for all  $\hat{z} \in Z_{\text{buf}}(k - t_{\text{rep}})$  do
27:          $Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup \{\hat{z}\}$ 
28:       end for
29:     end if
30:   end if
31: Return  $Z_k^{\text{ghost}}$ 

```

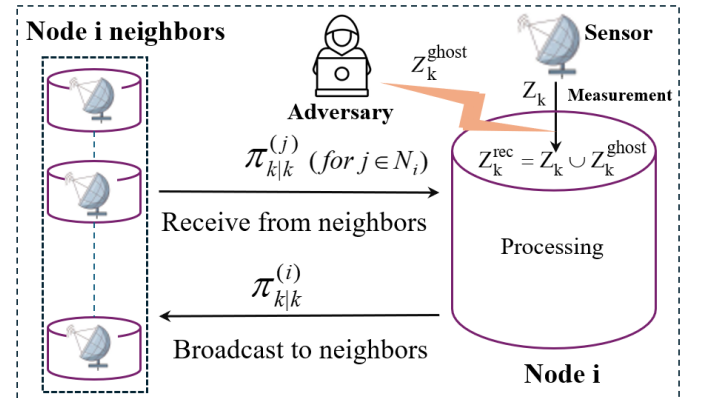


Fig. 1. Schematic of measurement-oriented ghost attacks on sensor i at the k^{th} time step.

a) *Communication phase*: In addition to the local sensor measurement data, each node i receives the multi-object prior densities from its neighboring nodes $j \in N_i$.

b) *Filtering phase*: Each node first performs a prediction step to compute predicted prior density $\pi_{k+1|k}^{(i)}$ as in Sec-

tion III. Before incorporating new measurements, correspondence between labels at node i and node j is established by defining the label mapping $\tau_{ij}: \mathbb{L}_i \rightarrow \mathbb{L}_j \cup \{\text{NULL}\}$, where each $\ell \in \mathbb{L}_i$ (for readability, the time index k has been omitted from $\mathbb{L}_i$ and $\mathbb{L}_j$) is assigned as

$$\tau_{ij}(\ell) = \begin{cases} \ell', & \text{if } \ell \text{ is associated with label } \ell' \in \mathbb{L}_j, \\ \text{NULL}, & \text{otherwise.} \end{cases} \quad (18)$$

The collection of all such mappings is denoted Γ^{ij} . The optimal mapping $\bar{\tau}_{ij}$ is obtained by solving

$$\begin{aligned} \bar{\tau}_{ij} &= \arg \min_{\tau_{ij} \in \Gamma^{ij}} J(\pi_{\tau_{ij}}^{(j)}, \pi_{\tau_{ij}}^{(i)}) \\ &= \arg \min_{\tau_{ij} \in \Gamma^{ij}} \left[D_{\text{KL}}(\pi_{\tau_{ij}}^{(j)} \parallel \pi_{\tau_{ij}}^{(i)}) \right. \\ &\quad \left. + D_{\text{KL}}(\pi_{\tau_{ij}}^{(i)} \parallel \pi_{\tau_{ij}}^{(j)}) \right], \end{aligned} \quad (19)$$

where $\pi_{\tau_{ij}}^{(j)}$ and $\pi_{\tau_{ij}}^{(i)}$ are the augmented densities at nodes j and i , respectively, based on the label-partitioning scheme in [39]. $D_{\text{KL}}(p \parallel q)$ is the Kullback–Leibler divergence [57], and

$$J(p, q) = D_{\text{KL}}(p \parallel q) + D_{\text{KL}}(q \parallel p) \quad (20)$$

is the Jeffreys divergence, a symmetric measure of dissimilarity between distributions [58].

Once these correspondences are known, the label-reliability factor for each label $\ell \in \mathbb{L}_i$ is computed as follows:

$$\bar{\alpha}_i(\ell) = \begin{cases} \frac{\sum_{j \in N_i} 1_{\mathbb{L}_j}(\bar{\tau}_{ij}(\ell))}{|N_i|}, & |N_i| \geq 1, \\ 1, & |N_i| = 0, \end{cases} \quad (21)$$

where

$$1_{\mathbb{L}_j}(\bar{\tau}_{ij}(\ell)) = \begin{cases} 1, & \bar{\tau}_{ij}(\ell) \in \mathbb{L}_j, \\ 0, & \text{otherwise,} \end{cases}$$

accounts for the existence of label ℓ in neighboring node j , i.e., the vote cast by node j for label ℓ . Here, $\bar{\alpha}_i(\ell) \in [0, 1]$ quantifies the confidence in track ℓ at node i by averaging how many neighbors also recognize that same label, thus reflecting consensus. Finally, the modified single-object likelihood at node i is defined as

$$\begin{aligned} \bar{g}_k^{(i)}(z_{\theta(\ell)} \mid x, \ell) &= \bar{\alpha}_i(\ell) g_{k, \text{true}}^{(i)}(z_{\theta(\ell)} \mid x, \ell) \\ &\quad + [1 - \bar{\alpha}_i(\ell)] g_{k, \text{ghost}}^{(i)}(z_{\theta(\ell)} \mid x, \ell), \end{aligned} \quad (22)$$

where $\theta \in \Theta_k$, and Θ_k is the set of all mappings $\theta: \mathbb{L} \rightarrow \{0, \dots, |Z_k^{\text{rec}}|\}$. In (22), $g_{k, \text{true}}^{(i)}(z_{\theta(\ell)} \mid x, \ell)$ denotes the original likelihood of a true object, whereas the ghost likelihood at sensor i is modeled as the small constant ε_i :

$$g_{k, \text{ghost}}^{(i)}(z_{\theta(\ell)} \mid x, \ell) = \varepsilon_i, \quad 0 < \varepsilon_i \ll 1.$$

A convenient choice is $\varepsilon_i = \frac{1}{|\text{FoV}_i|}$, where FoV_i denotes the field of view of sensor i which is assumed to be sufficiently large to encompass the region of interest. Since $\varepsilon_i \ll 1$, any ghost measurement carries negligible probability relative to true target returns. Substitution into (22) yields

$$\bar{g}_k^{(i)}(z_{\theta(\ell)} \mid x, \ell) = \bar{\alpha}_i(\ell) g_{k, \text{true}}^{(i)}(z_{\theta(\ell)} \mid x, \ell) + [1 - \bar{\alpha}_i(\ell)] \varepsilon. \quad (23)$$

In the ghost-only case ($\bar{\alpha}_i(\ell) = 0$), the likelihood reduces to

$$\bar{g}_k^{(i)}(z_{\theta(\ell)} \mid x, \ell) \approx \varepsilon \ll 1.$$

Note that the label-reliability factor attenuates the likelihoods of ghost targets by requiring neighboring nodes to corroborate each object's *presence*; without sufficient overlap in these observations, measurements of genuine targets may be discarded (i.e., $\alpha_i(\ell) \rightarrow 0$).

The ALARM measurement update then employs this consensus-weighted likelihood to produce a multi-object posterior that is robust against stealthy, measurement-oriented adversarial attacks.

Remark 3. ALARM rejects ghost measurements during the *initial stage* of the filtering process by leveraging votes from neighboring nodes. If node i is under a ghost attack but has enough neighbors not under attack to outvote the falsified measurements (see Remark 4), all ghost targets are effectively discarded before any state update occurs. While one could further enhance the robustness of the state estimate by fusing the filtered densities, that fusion step lies beyond the scope of this work.

B. λ -ALARM

To further mitigate ghost-target influence, a consensus-based *shrinkage criterion* leveraging neighboring votes is introduced. For each label $\ell \in \mathbb{L}_i$, the shrinkage indicator $\lambda_i(\ell)$ is defined as

$$\lambda_i(\ell) = \begin{cases} 1, & \sum_{j \in N_i} 1_{\mathbb{L}_j}(\bar{\tau}_{ij}(\ell)) \geq |N_i| - T_n, \\ 0, & \text{otherwise,} \end{cases} \quad (24)$$

where N_i denotes the set of neighbors of node i and T_n is the maximum number of tolerated missed detections among those neighbors. If $\lambda_i(\ell) = 1$, the original label-reliability factor is retained; if $\lambda_i(\ell) = 0$, it is set to zero, thereby removing potential ghost tracks. Therefore, the modified label-reliability factor is defined as

$$\hat{\alpha}_i(\ell) = \lambda_i(\ell) \bar{\alpha}_i(\ell). \quad (25)$$

Substituting (25) into the ALARM single-object likelihood (22) yields the λ -ALARM likelihood:

$$\begin{aligned} \bar{g}_k^{(i)}(z_{\theta(\ell)} \mid x, \ell) &= \hat{\alpha}_i(\ell) g_{k, \text{true}}^{(i)}(z_{\theta(\ell)} \mid x, \ell) \\ &\quad + [1 - \hat{\alpha}_i(\ell)] g_{k, \text{ghost}}^{(i)}(z_{\theta(\ell)} \mid x, \ell). \end{aligned} \quad (26)$$

The ALARM method and its stricter variant, λ -ALARM, comprise a generally applicable enhancement for any labeled RFS filtering algorithm. In this paper, λ -ALARM is implemented within the LMB filter for a multi-sensor environment. Henceforth, “ALARM” refers to LMB- λ -ALARM.

VI. IMPLEMENTATION DETAILS

The implementation of the proposed ALARM method using the LMB filter is detailed below, with the LMB filter efficiently executed by performing the joint prediction and update step proposed in [2], [53]. Existing multi-object filtering algorithms based on labeled multi-Bernoulli densities usually make use of either a particle or a Gaussian mixture representation. In

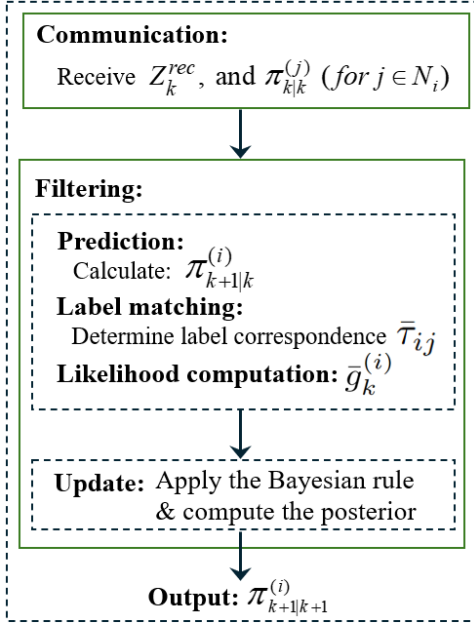


Fig. 2. Block diagram illustrating the ALARM filtering scheme at node i during time step k .

this work, we adopt a particle-based representation, wherein the kinematic state of each target is modeled using a set of weighted particles. These particles are processed using the Sequential Importance Resampling (SIR) particle filter framework for both prediction and update steps. At each node i , the state of the LMB filter at time k is represented as $\mathbf{X}_{k|k}^{(i)} = \{(r_{k|k}^{(\ell)}, p_{k|k}^{(\ell)}(x)) \mid \ell \in \mathbb{L}_i\}$, where $r_{k|k}^{(\ell)}$ is the existence probability of object ℓ , and $p_{k|k}^{(\ell)}(x)$ is its SPDF. Each SPDF is approximated by a set of particles:

$$p_{k|k}^{(\ell)}(x) \cong \sum_{m=1}^{v_{k|k}} w_{k|k}^{m,(\ell)} \delta(x - x_{k|k}^{m,(\ell)}), \quad (27)$$

where $x_{k|k}^{m,(\ell)}$ represents the particle states, $w_{k|k}^{m,(\ell)}$ are the corresponding weights, $v_{k|k}$ is the number of particles per each track, and $\delta(x)$ is the Dirac delta function, defined as:

$$\delta(x) \triangleq \lim_{\sigma \rightarrow 0} \frac{1}{|\sigma| \sqrt{\pi}} e^{-(x/\sigma)^2}. \quad (28)$$

More rigorously, the set of weighted particles $\{x_{k|k}^{m,(\ell)}, w_{k|k}^{m,(\ell)}\}_{m=1}^{v_{k|k}}$ constitutes a Monte Carlo approximation of the posterior density $p_{k|k}^{(\ell)}(x)$ if, for any bounded and unitless test function $\theta(x)$, the following approximation holds [18]:

$$\int \theta(x) p_{k|k}^{(\ell)}(x) dx \cong \sum_{m=1}^{v_{k|k}} w_{k|k}^{m,(\ell)} \theta(x_{k|k}^{m,(\ell)}). \quad (29)$$

Moreover, as the number of particles $v_{k|k}$ becomes arbitrarily large, the approximation converges to the true expected value:

$$\int \theta(x) p_{k|k}^{(\ell)}(x) dx = \lim_{v_{k|k} \rightarrow \infty} \sum_{m=1}^{v_{k|k}} w_{k|k}^{m,(\ell)} \theta(x_{k|k}^{m,(\ell)}).$$

The convergence is guaranteed by the law of large numbers, establishing the validity of the particle-based representation for Bayesian filtering in the limit of large sample sizes.

New targets are introduced into the system via a birth process, which accounts for the spontaneous appearance of new objects in the scene. At each time step, a set of N_B birth components is generated, each initialized with an existence probability $r_{k|k}^{(\ell_B)}$, where $B = \{1, \dots, N_B\}$. The spatial distribution of newly born targets is modeled using a Gaussian mixture model, where the birth particles are sampled from a mixture of Gaussian components, each characterized by a mean, covariance, and an associated weight.

The birth process is given by:

$$\{x_{k|k}^{m,(\ell_B)}\}_{m=1}^v \sim \sum_{b=1}^{N_c} w_b \mathcal{N}(m_b, P_b), \quad (30)$$

where v denotes the number of particles generated for each birth component, N_c denotes the number of Gaussian components, $x_{k|k}^{m,(\ell_B)}$ represents a sampled particle for a birth track ℓ_B , w_b is the weight of the b -th Gaussian component, and $\mathcal{N}(m_b, P_b)$ is a Gaussian distribution with mean m_b and covariance P_b . The corresponding birth weights are initialized uniformly as:

$$w_{k|k}^{m,(\ell_B)} = \frac{1}{v}, \quad m = 1, \dots, v. \quad (31)$$

The particles and their associated weights evolve according to the motion model within the *prediction step* of the SIR framework. In general, for each existing track ℓ , the predicted particles $\{x_{k+1|k}^{m,(\ell)}\}_{m=1}^{v_{k+1|k}}$ are generated from an importance-sampling density. In this work we adopt the bootstrap approach, in which the Markov density (“dynamic prior”) is used as the importance-sampling density [18]. Hence, particles are propagated according to the dynamic model:

$$x_{k+1|k}^{m,(\ell)} \sim \mathcal{N}(F x_{k|k}^{m,(\ell)}, Q), \quad (32)$$

where F is the state transition matrix and Q is the process noise covariance matrix. The predicted weights are then calculated and normalized as:

$$w_{k+1|k}^{m,(\ell)} = p_S(x_{k|k}^{m,(\ell)}) \times w_{k|k}^{m,(\ell)}, \quad (33)$$

$$\tilde{w}_{k+1|k}^{m,(\ell)} = \frac{w_{k+1|k}^{m,(\ell)}}{\sum_{m=1}^{v_{k+1|k}} w_{k+1|k}^{m,(\ell)}}, \quad (34)$$

where p_S is the probability of survival. Also, the existence probability is evolved by:

$$r_{k+1|k}^{(\ell)} = \sum_{m=1}^{v_{k+1|k}} \tilde{w}_{k+1|k}^{m,(\ell)} \times r_{k|k}^{(\ell)}. \quad (35)$$

Note that $r_{k+1|k}^{(\ell)}$ can be interpreted as the average probability of survival $\bar{p}_S^{(\ell)}$ for each track. Therefore:

$$\bar{p}_S^{(\ell)} = r_{k+1|k}^{(\ell)}. \quad (36)$$

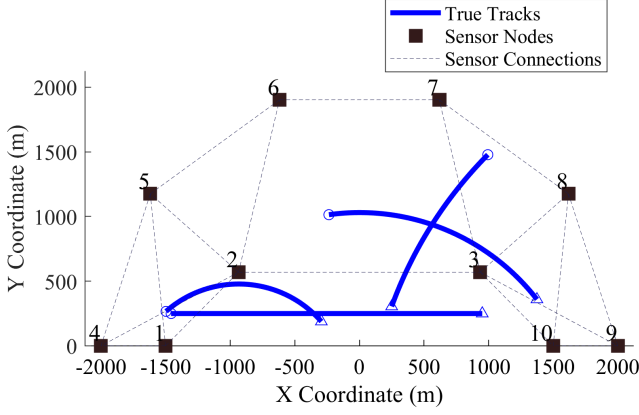


Fig. 3. Ground truths and the sensor network layout. Start/Stop positions given by $\circ/\triangle$.

Following the prediction step, the particle weights are updated using the average measurement likelihood (26), and normalized as:

$$w_{k+1|k+1}^{m,(\ell)} = \bar{p}_S^{(\ell)} p_D(x_{k+1|k}^{m,(\ell)} | \bar{g}_k^{(i)}(z_{\theta(\ell)} | x_{k+1|k}^{m,(\ell)}, \ell) \times \tilde{w}_{k+1|k}^{m,(\ell)} \quad (37)$$

$$\tilde{w}_{k+1|k+1}^{m,(\ell)} = \frac{w_{k+1|k+1}^{m,(\ell)}}{\sum_{m=1}^{v_{k+1|k+1}} w_{k+1|k+1}^{m,(\ell)}}, \quad (38)$$

where p_D is the detection probability, and $j \in N_i$ denotes the neighboring nodes of node i . Note that the particles evolve solely based on the motion model, with the update step applied only to the weights. In other words, $x_{k+1|k+1}^{m,(\ell)} = x_{k+1|k}^{m,(\ell)}$.

VII. SIMULATIONS

In this section, the performance of the proposed method is evaluated using simulation experiments conducted on a virtual distributed sensor network implemented in the MATLAB 2024b environment for multi-target tracking.

A. Simulation setup

This work considers a nonlinear multi-target tracking scenario, as described in [24], with a total of four targets. The state of each target at time step k , represented as $\mathbf{x}_k = [\tilde{\mathbf{x}}_k^T \ \omega_k]^T$, consists of the planar position and velocity $\tilde{\mathbf{x}}_k = [p_{x,k}, \dot{p}_{x,k}, p_{y,k}, \dot{p}_{y,k}]^T$, along with the turn rate ω_k . The target population is dynamic, with the number of targets varying over time due to the processes of births and deaths. Sensor returns are bearings and range vectors of the form $z_k = [\theta_k, r_k]^T$ on the half disc of radius 2000 m, and the observations are subject to missed detections and clutter. To track the targets, a sensor network consisting of $|V| = 10$ sensor nodes is deployed in the surveillance area. Each node can communicate with its neighboring nodes if a connection exists, allowing them to exchange their local multi-object states. Fig. 3 shows the ground truths and the sensor network layout. All sensors are configured with an identical 2000 m FoV, so their placement does not affect the outcome.

Individual targets follow a coordinated turn model [24] with transition density $f_{k|k-1}(x' | x_k) = \mathcal{N}(x'; m(x_k), Q)$, where

$$m(x_k) = [F(\omega_k) \tilde{x}_k]^T, \quad Q = \text{diag}([\sigma_w^2 G G^T, \sigma_u^2]),$$

$$F(\omega) = \begin{bmatrix} 1 & \frac{\sin \omega \Delta}{\omega} & 0 & -\frac{1 - \cos \omega \Delta}{\omega} \\ 0 & \cos \omega \Delta & 0 & -\sin \omega \Delta \\ 0 & \frac{1 - \cos \omega \Delta}{\omega} & 1 & \frac{\sin \omega \Delta}{\omega} \\ 0 & \sin \omega \Delta & 0 & \cos \omega \Delta \end{bmatrix}, \quad (39)$$

$$G^T = \begin{bmatrix} \frac{\Delta^2}{2} & 0 \\ 0 & \frac{\Delta^2}{2} \\ \Delta & 0 \\ 0 & \Delta \end{bmatrix}, \quad (40)$$

$\Delta = 1$ s, and $\sigma_w = 15$ m/s² is the standard deviation of the process noise, and $\sigma_u = \frac{\pi}{180}$ rad/s is the standard deviation of the turn rate noise.

The survival probability for targets is $p_{S,k}(x) = 0.99$, for all sensor nodes. The birth process at each node follows a labeled Poisson RFS [24] with intensity

$$\gamma_k(x) = \sum_{i=1}^4 w_\gamma^{(i)} \mathcal{N}(x; m_\gamma^{(i)}, P_\gamma), \quad (41)$$

where $w_\gamma^{(1)} = w_\gamma^{(2)} = 0.02$, $w_\gamma^{(3)} = w_\gamma^{(4)} = 0.03$,

$$m_\gamma^{(1)} = [-1500, 0, 250, 0, 0]^T, m_\gamma^{(2)} = [-250, 0, 1000, 0, 0]^T,$$

$$m_\gamma^{(3)} = [250, 0, 750, 0, 0]^T, m_\gamma^{(4)} = [1000, 0, 1500, 0, 0]^T,$$

and $P_\gamma = \text{diag}([50, 50, 50, 50, 6(\pi/180)^2])^2$.

If detected, each target produces a noisy bearing and range measurement $z = [\theta, r]^T$ with likelihood

$$g_k(z|x) = \mathcal{N}(z; \mu(x), R), \quad (42)$$

where $\mu(x) = [\arctan(p_x/p_y), \sqrt{p_x^2 + p_y^2}]$ and $R = \text{diag}([\sigma_\theta^2, \sigma_r^2]^T)$, with $\sigma_\theta = \frac{\pi}{180}$ rad and $\sigma_r = 5$ m. For a fair comparison between ALARM and standard filtering, our simulation assumes that all nodes adhere to a same detection probability rule. The probability of detection is state-dependent and is given by $p_{D,k}(x) \propto \mathcal{N}(x; [0, 0], \text{diag}([2000, 2000])^2)$, which reaches its peak value of 0.98 at the origin and gradually decreases to its lowest value at the edge of the surveillance region.

Clutter follows a Poisson RFS with intensity $\kappa_k(z) = \lambda_c U(Z)$, where $\lambda_c = 1.59 \times 10^{-3}$ (rad/m)⁻¹ and $U(Z)$ denotes a uniform density on the observation region, giving an average of 10 clutter points per scan.

B. Computational Considerations

The filter implementation follows the efficient joint prediction and update methodology proposed in [2] where instead of performing two truncations, only one truncation is executed per iteration, reducing computational costs. Additionally, the truncation process is enhanced using Gibbs sampling, which is more efficient than the previous ranked assignment methods. To manage computational complexity, the number of new

components calculated and stored in each forward propagation is set to be proportional to the weight of the original component. Components with very low weights (below 10^{-3}) are discarded, and the total number of components is truncated to a maximum of 100. The pseudo-code for the filter implementation is provided in Appendix B.

C. Attack configuration

As described in Algorithm 1, ghost attacks can occur in three primary scenarios. In *deception*, new targets are injected into the system, mimicking the motion model of real objects to mislead the tracking system. In a *replay* attack, previously recorded measurements are retransmitted into the system, creating confusion for the tracker. Finally, *delay* attacks involve injecting a large volume of ghost targets to overwhelm the system, exhausting computational resources and causing significant delays or potential system failure (DoS).

1) *Scenario 1 (Deception)*: In the simulations conducted for this scenario, the adversary is assumed to be aware of the transition model and executes stealthy attacks by injecting measurements that closely resemble the normal target's motion, in order to deceive the filtering process. Following Algorithm 1, ghost attack measurements are generated using the transition model described in (39). The number of ghost objects in this scenario includes 4, 6, 12, and 16, corresponding to ghost densities of 50%, 60%, 75%, and 80%, respectively. The birth and death times, initial positions, and turn rates of the ghost targets are detailed in Table II.

Remark 4. In a network where each node has d neighbors, a standard result from resilient consensus theory [59] states that each node can locally tolerate up to

$$f = \left\lfloor \frac{d-1}{2} \right\rfloor \quad (43)$$

malicious neighbors. Here, since d is 3 or 4 in our setting (see Fig. 3), we have $f = \left\lfloor \frac{4-1}{2} \right\rfloor = \left\lfloor \frac{3-1}{2} \right\rfloor = 1$. Hence, the maximum number of compromised neighbors a node can tolerate is $f = 1$ under majority-based or robust-fusion protocols that require at least $2f + 1$ neighbors to outvote malicious inputs. To inject ghost targets, we selected two nodes with a degree of 4 (nodes 5, and 8), ensuring that each node in the network remained connected to at least one neighboring node affected by the attack. Also, we set the threshold T_n in (24) to 4, which represents the minimum number of neighboring nodes required to unanimously vote for a (ghost) measurement likelihood to be strictly set to zero.

2) *Scenario 2 (Replay)*: In this scenario, the attacker records the measurements of the four true targets and injects them back into the target node after $t_{rep} = 10$ time steps. To simulate this scenario, it is assumed that the attacker is aware of the true measurements and the measurement noise covariance. The attacker then generates ghost measurements with added noise, sampled from a normal distribution. This process is repeated multiple times, with the attacker retransmitting the ghost measurements for all four targets each time. The performance of the ALARM method is evaluated by targeting similar nodes in Scenario 1.

TABLE II
GHOST TARGET SPECIFICATIONS IN SIMULATION ($w_{turn} = \frac{2\pi}{180}$).

Target ID	t_{birth}	t_{death}	x Position	y Position	Turn Rate
1	10	100	1003	1488	$w_{turn}/8$
2	10	100	-255	1011	$-w_{turn}/3$
3	10	100	-1507	256	$-w_{turn}/2$
4	10	66	-1499	251	0
5	20	80	246	738	$w_{turn}/4$
6	20	80	246	738	$w_{turn}/4$
7	40	100	-242	993	$w_{turn}/2$
8	40	100	1000	1500	$w_{turn}/4$
9	40	80	250	750	$-w_{turn}/4$
10	40	100	-242	993	$w_{turn}/2$
11	40	100	1001	1501	$w_{turn}/4$
12	40	80	251	752	$-w_{turn}/4$
13	60	100	1000	1500	$-w_{turn}/4$
14	60	100	250	750	$w_{turn}/4$
15	60	100	999	1498	$-w_{turn}/4$
16	60	100	252	753	$w_{turn}/4$

3) *Scenario 3 (Delay)*: Ghost attacks not only degrade the accuracy of estimation but also introduce additional computational costs and delays, potentially compromising the real-time performance of the system. In a real-time system, correctness depends not only on the logical accuracy of computations but also on their timing. Such systems must produce correct outputs within a specified time frame. However, standard filtering methods lack mechanisms to mitigate stealthy ghost attacks. Therefore, an attacker can exploit this vulnerability by injecting a high volume of ghost measurements, causing significant delays, and in severe cases, leading to a DoS attack. To simulate a high-volume ghost attack, Scenario 1 (featuring 16 ghost objects) is combined with Scenario 2 (involving 4 ghost objects). The ALARM method's performance is then evaluated by targeting nodes similar to those in Scenario 1. Additionally, the average execution time and cardinality statistics for both ALARM and the standard filtering method are compared.

D. Results

In this section, the tracking evaluation results for both the standard and ALARM filtering methods are presented for each scenario. To assess the accuracy of the tracking performance, the Optimal Sub-Pattern Assignment (OSPA) metric is used, which compares two finite sets of points in multi-object filtering. It measures the similarity between two sets, X and Y , often representing estimated and ground-truth object states, respectively. While *OSPA* measures the error between true and estimated states at a single time step, *OSPA*⁽²⁾ extends this by measuring the error between true and estimated tracks over a window of time steps. For a detailed explanation of the *OSPA* and *OSPA*⁽²⁾ metrics, refer to Appendix C.

The evaluation results for all attack scenarios, based on the *OSPA* and *OSPA*⁽²⁾ distance error metrics, are summarized in

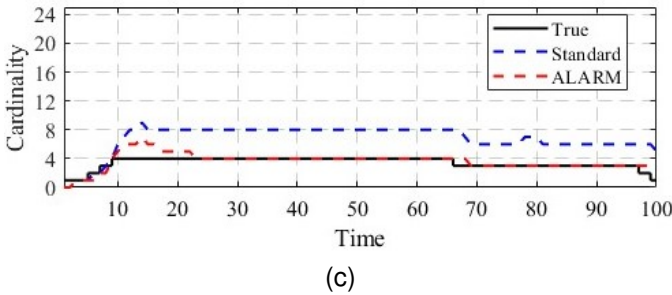
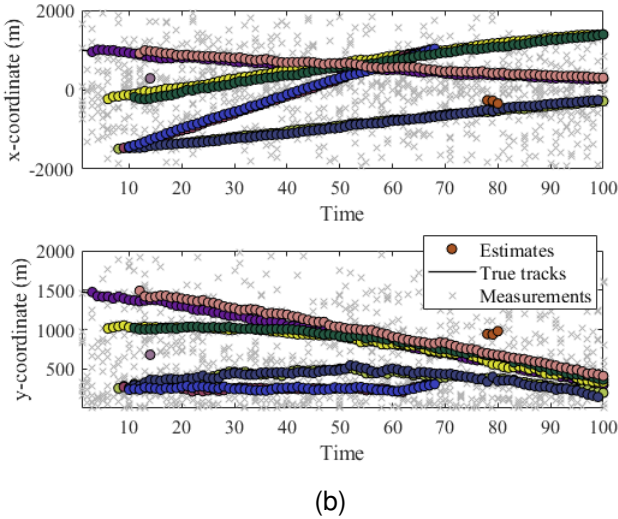
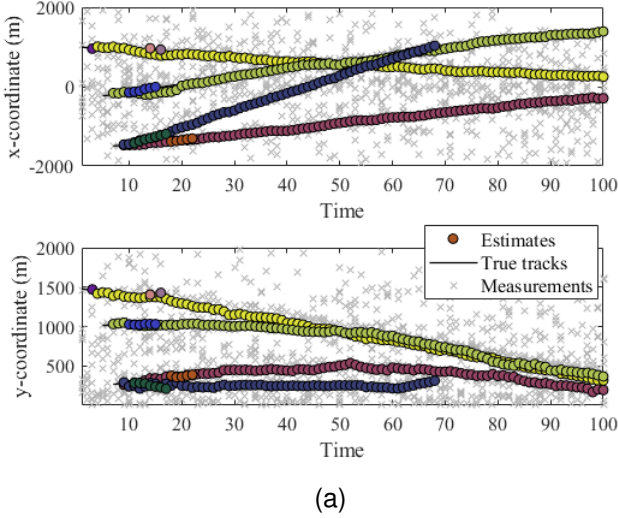


Fig. 4. Tracking performance of ALARM at the targeted nodes (nodes 5, and 8) and standard filtering under a 50% density ghost attack: (a) tracking results for ALARM, (b) tracking results for standard filtering, and (c) cardinality statistics comparison.

Table III. The table compares the performance of the ALARM and standard filtering methods across deception, replay, and delay attack scenarios, evaluated at varying ghost densities. Ghost density represents the proportion of false targets (ghosts) injected into the system, expressed as a percentage. For the ALARM method, the table presents the corresponding average

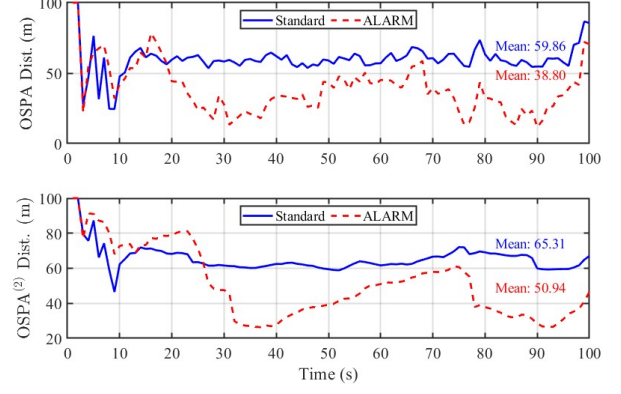


Fig. 5. Comparison of $OSPA$ and $OSPA^{(2)}$ distance error metrics for standard and ALARM (at the targeted nodes) methods under the 50% density ghost attack (window=10, cut-off=100 meters, and order=1).

$OSPA$ and $OSPA^{(2)}$ values when two nodes (nodes 5 and 8) are simultaneously under attack. Additionally, the attack scenario for the standard method is always simulated with a single node under attack.

In the deception scenario, ALARM maintains consistently superior performance over four levels of ghost density. At 50% density (4 ghosts), ALARM achieves an $OSPA$ distance of 27.21 m, outperforming the standard filter's 59.85 m, and also shows a lower $OSPA^{(2)}$ distance (36.53 m vs. 65.31 m). As the density increases to 60% (6 ghosts), ALARM extends its advantage with 30.96 m versus 65.36 m in $OSPA$ distance and 39.82 m versus 71.43 m in $OSPA^{(2)}$ distance. At 75% density (12 ghosts), ALARM records 32.64 m compared to the standard filter's 73.10 m, with a corresponding $OSPA^{(2)}$ distance of 42.17 m versus 77.89 m. Even at 80% density (16 ghosts), ALARM sustains its advantage by registering 32.89 m against the standard filter's 75.17 m, alongside $OSPA^{(2)}$ results of 42.38 m versus 79.65 m, thus illustrating ALARM's robust resilience to increasing ghost interference.

For the 50% ghost density scenario, the tracking estimates (along with the corresponding cardinality statistics) and the $OSPA/OSPA^{(2)}$ evaluations for both ALARM and the standard filtering method are presented in Fig. 4 and Fig. 5, respectively. Because nodes 5 and 8 have identical features in our simulations, their tracking results are the same; hence, only one node's results are shown. The results show that when ghosts are injected into targeted nodes at the 10th time step, ALARM initially recognizes them as valid tracks but subsequently rejects these ghost measurements after a few time steps, whereas the standard filtering method continues to treat all ghost targets as valid. These findings demonstrate that ALARM effectively rejects ghost measurements, yielding more accurate and consistent tracking estimates, superior cardinality statistics, and lower $OSPA$ and $OSPA^{(2)}$ errors.

In the replay attack scenario, where previously recorded ghost measurements are reinjected into the system after 10 time steps, ALARM achieves an $OSPA$ distance error of 24.89 m, compared to 58.47 m for the standard filtering approach. This suggests that replay attacks, which more

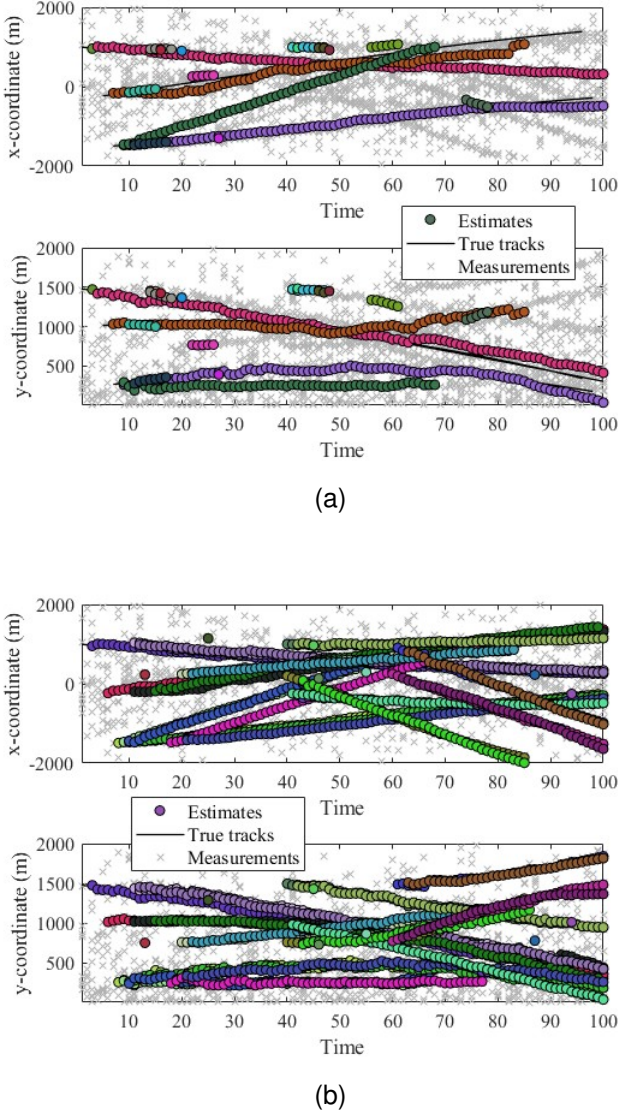


Fig. 6. Tracking performance of ALARM at the targeted nodes (nodes 5, and 8) and standard filtering under a 83% density ghost attack: (a) tracking estimates for ALARM, (b) tracking estimates for standard filtering.

closely resemble real objects, pose a greater challenge for mitigation. A similar trend is observed with $OSPA^{(2)}$ (34.37 m vs 64.51 m).

The delay attack scenario combines deception with 16 ghost targets and replay with 4 ghost targets, introducing a total of 20 ghost targets to impose additional computational overhead on the system. Under this condition, ALARM achieves an outstanding $OSPA$ distance error of 32.38 m, compared to 79.51 m for the standard filtering method. A similar behavior is also observed with $OSPA^{(2)}$ (41.46 m vs 84.09 m). Figure 6 presents the tracking estimates for both ALARM and the standard filtering method in this scenario. It can be seen that the standard method mistakenly detects all ghost targets, while the high-volume ghost measurement injection causes some short-term misdetections and slight disruptions in detecting the true targets under ALARM. Moreover, the execution times for both methods under the delay attack scenario is presented

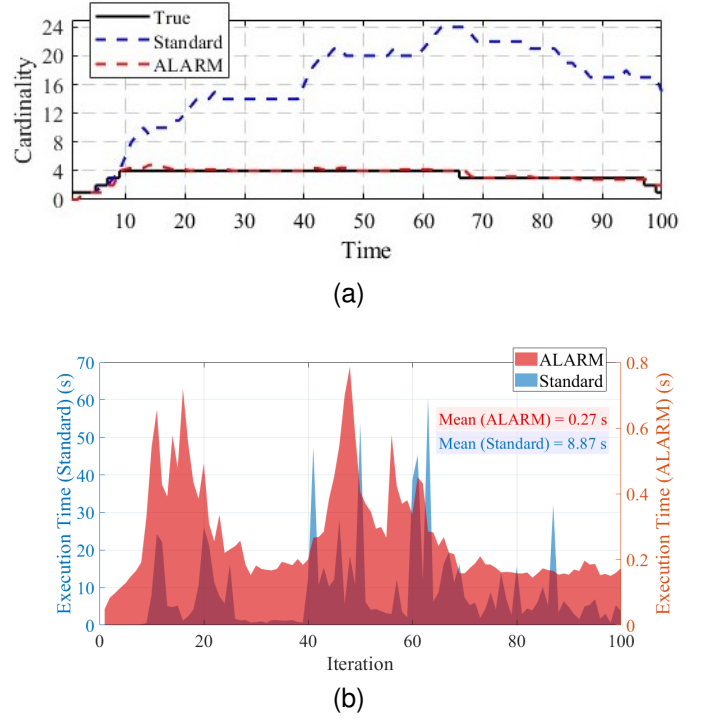


Fig. 7. (a) Comparison of average cardinality statistics and (b) average execution time per node in the delay attack scenario with 20 ghost targets (the average execution time plots use **different time scales**).

in Fig. 7. The figure illustrates how the average execution time per iteration increases when ghost tracks are detected by the standard filter. The average execution time of the standard filtering method (8.87 s) in this case is substantially higher than that of the ALARM method (0.27 s). It is evident that, with higher attack densities, the standard filtering node would experience significant slowdowns and could potentially be rendered inoperable, leading to a DoS scenario. These measurements were taken on a machine with an Intel® Core™ i7-8700 CPU operating at 3.20 GHz.

Considering the large volume of ghost targets (20 out of 24), ALARM demonstrates:

- 1) Significantly better accuracy than the standard filtering method (lower $OSPA$ and $OSPA^{(2)}$ errors),
- 2) Dismissal of nearly all ghost targets after only a few time steps, and
- 3) Significantly reduced execution time.

The increase in execution time during filtering simulation could be attributed to the complexity of the data and the increased number of hypotheses. As the number of measurements and objects grows, the algorithm's complexity increases, particularly during the update step. This can result in longer processing times for prediction and correction. Additionally, if the number of possible object tracks or hypotheses increases, the computation time required for hypothesis pruning or measurement association also grows, further impacting execution time. Since the ALARM filtering method is more robust and rejects more ghost measurements, it reduces the need to process false measurements related to ghost targets, thereby lowering the computational cost.

TABLE III
COMPARISON OF ALARM AND STANDARD FILTERING METHODS USING $OSPA$ AND $OSPA^{(2)}$ (WINDOW=10, CUT-OFF=100 METERS, AND ORDER=1) METRICS.

Attack Type	Ghost Density (%)	Avg. $OSPA$ Dist. (m)↓		Avg. $OSPA^{(2)}$ Dist. (m)↓	
		ALARM	Standard	ALARM	Standard
Deception	50 (4 ghosts)	27.21	59.85	36.53	65.31
	60 (6 ghosts)	30.96	65.36	39.82	71.43
	75 (12 ghosts)	32.64	73.10	42.17	77.89
	80 (16 ghosts)	32.89	75.17	42.38	79.65
Replay	50 (4 ghosts)	24.89	58.47	34.37	64.51
Delay	83 (20 ghosts)	32.38	79.51	41.46	84.09

In summary, the results show that the ALARM effectively mitigates the impact of ghost attacks, consistently achieving lower error metrics even under high attack density. Additionally, it reduces computational overhead, making it well-suited for real-time applications. These outcomes emphasize the method's ability to enhance the integrity of the system by accurately rejecting false measurements and its accessibility by ensuring timely and reliable operation, even under high attack intensities. This makes ALARM a practical and efficient solution, particularly for large-scale networks susceptible to attacks.

VIII. CONCLUSIONS

This paper addresses stealthy measurement-oriented adversarial attacks in multi-sensor, multi-object tracking systems, which are crucial for maintaining system integrity and accessibility. A robust filtering algorithm, ALARM, is proposed to mitigate the effects of ghost attacks by effectively rejecting false measurements and preserving accurate object tracking. The evaluation results show that ALARM significantly reduces computational overhead, with execution times substantially shorter than those of the standard filtering method. This efficiency underscores its suitability for real-time applications where timely processing is essential. The reduction in computational cost also helps mitigate the risk of denial-of-service (DoS) vulnerabilities, which can result from increased processing demands during high-density ghost attacks. These features make ALARM a practical and efficient solution, particularly for large-scale networks susceptible to attacks. Since the proposed filtering framework is general in terms of the filtering method, future research could extend its application to filters other than LMB and investigate its socio-economic impact in real-world transportation scenarios [60], [61].

APPENDIX A δ -GLMB FILTER

A δ -generalized labeled multi-Bernoulli RFS with state space $\mathbb{X}$ and (discrete) label space $\mathbb{L}$ is a special case of a generalized labeled multi-Bernoulli with [24]

$$\begin{aligned}\mathbb{C} &= \mathcal{F}(\mathbb{L}) \times \Xi, \\ w^{(c)}(L) &= w^{(I, \xi)}(I_{\Xi}) = w^{(I, \xi)}\delta_I(L), \\ p^{(c)} &= p^{(I, \xi)} = p^{(\xi)},\end{aligned}$$

where Ξ is a discrete space, i.e., it is distributed according to

$$\pi(\mathbf{X}) = \Delta(\mathbf{X}) \sum_{(I, \xi) \in \mathcal{F}(\mathbb{L}) \times \Xi} w^{(I, \xi)} \delta_I(\mathcal{L}(\mathbf{X})) \left[p^{(\xi)} \right]^{\mathbf{X}} \quad (44)$$

The δ -generalized labeled multi-Bernoulli (δ -GLMB) framework represents either the multi-object prediction density or the multi-object filtering density at time k . At this time, each $I \in \mathcal{F}(\mathbb{L})$ corresponds to a set of track labels, while $\xi \in \Xi$ encodes a history of association maps. Specifically, in the prediction density, ξ represents the sequence of association maps up to time $k-1$, i.e., $\xi = (\theta_1, \dots, \theta_{k-1})$, where each θ_j is an association map at time j that links track labels to measurement indices. Conversely, in the filtering density, ξ extends to time k , meaning $\xi = (\theta_1, \dots, \theta_k)$. The pair (I, ξ) forms a *hypothesis*, indicating that the set of tracks I follows the measurement association history ξ . The weight $w^{(I, \xi)}$ quantifies the probability of this hypothesis, while the function $p^{(\xi)}(\cdot, \ell)$ specifies the kinematic state density of track ℓ under the given association history ξ . However, not all hypotheses are valid, as certain pairs (I, ξ) may be inconsistent; such infeasible hypotheses are assigned a weight of zero.

APPENDIX B LMB FILTERING ALGORITHM IMPLEMENTATION

An overview of the LMB implementation is presented in Algorithm 2.

APPENDIX C $OSPA^{(2)}$ METRIC

Beard et al. [36], [62] extended the $OSPA$ [63] metric to evaluate multi-target tracking performance by introducing a new metric called $OSPA^{(2)}$ [64]. The base distance used in $OSPA^{(2)}$ is derived from the $OSPA$ -based distance. While $OSPA$ measures the error between true and estimated states at a single time step, $OSPA^{(2)}$ measures the error between true and estimated tracks over a window of time steps. The $OSPA^{(2)}$ metric, $\check{d}_{p,q}^{(c)}(X, Y; w)$, is defined as follows [62], equation (4):

$$\begin{aligned}\check{d}_{p,q}^{(c)}(X, Y; w) &= \left(\frac{1}{n} \left(\min_{\pi \in \Pi_n} \sum_{i=1}^m \check{d}_q^{(c)}(f(i), g(\pi(i)), w)^p \right) \right. \\ &\quad \left. + c^p(n-m) \right)^{1/p}.\end{aligned}$$

Algorithm 2 Pseudo-Code for Efficient LMB Filtering

```

1: Initialization:
2: Set filter parameters (filter);
3: Initialize estimation storage (est);
4: Initialize track table (tt);
5: Determine the total number of simulations (K);
6: Filtering Process:
7: for  $k = 1$  to meas.K do
8:   A. Prediction and Update:
9:   Compute GLMB posterior:
10:   $\text{glmb} \leftarrow \text{PREDICTUPDATE}(\text{tt}, \text{filter}, \text{meas}[k]);$ 
11:   B. Convert and Prune:
12:   Convert to LMB:
13:    $\text{tt} \leftarrow \text{GLMBtoLMB}(\text{glmb});$ 
14:   Prune track table:
15:    $\text{tt} \leftarrow \text{PRUNE}(\text{tt}, \text{filter});$ 
16:   C. State Extraction and Logging:
17:   Extract estimates:
18:    $\text{est}[k] \leftarrow \text{EXTRACTSTATE}(\text{tt});$ 
19:   Log diagnostics:
20:    $\text{LOGDIAGNOSTICS}(\text{tt}, k, \text{est}, \text{filter});$ 
21: end for
22: Output:
23: Return est;

```

where c is the cutoff, p is the order, q is the order of the base distance, and w is a collection of convex weights. The term $\tilde{d}_q^{(c)}(f_S, f_T; w)$ represents the time-averaged OSPA between two tracks f_S and f_T for all times $t \in \{1, \dots, K\}$, where $\sum_{t \in T} w(t) = 1$ and q is a positive integer. The formula is given by [62], equation (2):

$$\tilde{d}_q^{(c)}(f_S, f_T; w) = \left(\sum_{t=1}^K \left[w(t) d^{(c)}(\{f_S(t)\}, \{f_T(t)\})^q \right] \right)^{1/q}.$$

REFERENCES

- [1] H. V. Nguyen, T. T. D. Nguyen, C. Shim, and M. Anuar, "The smooth trajectory estimator for lmb filters," in *2023 12th International Conference on Control, Automation and Information Sciences (ICCAIS)*, 2023, pp. 115–120.
- [2] B.-N. Vo, B.-T. Vo, and H. G. Hoang, "An efficient implementation of the generalized labeled multi-bernoulli filter," 1975.
- [3] A. Ghorbani, V. Ghorbani, M. Nazari-Heris, and S. Asadi, "Data assimilation for agent-based models," *Mathematics*, vol. 11, no. 20, p. 4296, 2023.
- [4] B. Ma, Z. Ren, and W. Cheng, "Traffic routing-based computation offloading in cybertwin-driven internet of vehicles for v2x applications," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 5, pp. 4551–4560, 2021.
- [5] H. V. Nguyen, M. Chesser, L. P. Koh, S. H. Rezatofghi, and D. C. Ranasinghe, "Trackerbots: Autonomous unmanned aerial vehicle for real-time localization and tracking of multiple radio-tagged animals," *Journal of Field Robotics*, vol. 36, no. 3, pp. 617–635, 2019.
- [6] B. Benfold and I. Reid, "Stable multi-target tracking in real-time surveillance video," in *CVPR 2011*. IEEE, 2011, pp. 3457–3464.
- [7] T. Rathnayake, R. Hoseinnezhad, R. Tennakoon, and A. Bab-Hadiashar, "Labeled multi-bernoulli tracking for industrial mobile platform safety," in *2017 IEEE International Conference on Mechatronics (ICM)*. IEEE, 2017, pp. 393–398.
- [8] T. Rathnayake, R. Tennakoon, A. Khodadadian Gostar, A. Bab-Hadiashar, and R. Hoseinnezhad, "Information fusion for industrial mobile platform safety via track-before-detect labeled multi-bernoulli filter," *Sensors*, vol. 19, no. 9, p. 2016, 2019.
- [9] R. Hoseinnezhad, B.-N. Vo, and B.-T. Vo, "Visual tracking in background subtracted image sequences via multi-bernoulli filtering," *IEEE Transactions on Signal Processing*, vol. 61, no. 2, pp. 392–397, 2012.
- [10] T. Rathnayake, A. K. Gostar, R. Hoseinnezhad, and A. Bab-Hadiashar, "Labeled multi-bernoulli track-before-detect for multi-target tracking in video," in *2015 18th International Conference on Information Fusion (Fusion)*. IEEE, 2015, pp. 1353–1358.
- [11] D. Stadler and J. Beyerer, "Improving multiple pedestrian tracking by track management and occlusion handling," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 10 958–10 967.
- [12] S. H. Rezatofghi, S. Gould, B. T. Vo, B.-N. Vo, K. Mele, and R. Hartley, "Multi-target tracking with time-varying clutter rate and detection profile: Application to time-lapse cell microscopy sequences," *IEEE transactions on medical imaging*, vol. 34, no. 6, pp. 1336–1348, 2015.
- [13] J. Liu, M. Liang, J. Ma, L. Jiang, H. Chu, C. Guo, J. Yu, Y. Zong, and M. Wan, "Microbubble tracking based on partial smoothing-based adaptive generalized labelled multi-bernoulli filter for super-resolution imaging," *Ultrasonics*, vol. 145, p. 107455, 2025.
- [14] A. K. Gostar, T. Rathnayake, R. Tennakoon, A. Bab-Hadiashar, G. Battistelli, L. Chisci, and R. Hoseinnezhad, "Centralized cooperative sensor fusion for dynamic sensor network with limited field-of-view via labeled multi-bernoulli filter," *IEEE Transactions on Signal Processing*, vol. 69, pp. 878–891, 2021.
- [15] Y. Bar-Shalom, F. Daum, and J. Huang, "The probabilistic data association filter," *IEEE Control Systems Magazine*, vol. 29, no. 6, pp. 82–100, 2009.
- [16] I. J. Cox and S. L. Hingorani, "An efficient implementation of reid's multiple hypothesis tracking algorithm and its evaluation for the purpose of visual tracking," *IEEE Transactions on pattern analysis and machine intelligence*, vol. 18, no. 2, pp. 138–150, 1996.
- [17] R. Mahler, *Statistical multisource-multitarget information fusion*. Artech, 2007.
- [18] R. P. Mahler, *Advances in statistical multisource-multitarget information fusion*. Artech House, 2014.
- [19] —, "Multitarget bayes filtering via first-order multitarget moments," *IEEE Transactions on Aerospace and Electronic systems*, vol. 39, no. 4, pp. 1152–1178, 2003.
- [20] B.-T. Vo, B.-N. Vo, and A. Cantoni, "Analytic implementations of the cardinalized probability hypothesis density filter," *IEEE transactions on signal processing*, vol. 55, no. 7, pp. 3553–3567, 2007.
- [21] —, "The cardinality balanced multi-target multi-bernoulli filter and its implementations," *IEEE Transactions on Signal Processing*, vol. 57, no. 2, pp. 409–423, 2008.
- [22] S. Reuter, B.-T. Vo, B.-N. Vo, and K. Dietmayer, "The labeled multi-bernoulli filter," *IEEE Transactions on Signal Processing*, vol. 62, no. 12, pp. 3246–3260, 2014.
- [23] B.-N. Vo, B.-T. Vo, and D. Phung, "Labeled random finite sets and the bayes multi-target tracking filter," *IEEE Transactions on Signal Processing*, vol. 62, no. 24, pp. 6554–6567, 2014.
- [24] B.-T. Vo and B.-N. Vo, "Labeled random finite sets and multi-object conjugate priors," *IEEE Transactions on Signal Processing*, vol. 61, no. 13, pp. 3460–3475, 2013.
- [25] C. Shim, B.-T. Vo, B.-N. Vo, J. Ong, and D. Moratuwage, "Linear complexity gibbs sampling for generalized labeled multi-bernoulli filtering," *IEEE Transactions on Signal Processing*, vol. 71, pp. 1981–1994, 2023.
- [26] T. T. D. Nguyen, B.-N. Vo, B.-T. Vo, D. Y. Kim, and Y. S. Choi, "Tracking cells and their lineages via labeled random finite sets," *IEEE Transactions on Signal Processing*, vol. 69, pp. 5611–5626, 2021.
- [27] D. S. Bryant, B.-T. Vo, B.-N. Vo, and B. A. Jones, "A generalized labeled multi-bernoulli filter with object spawning," *IEEE Transactions on Signal Processing*, vol. 66, no. 23, pp. 6177–6189, 2018.
- [28] F. Papi, B.-N. Vo, B.-T. Vo, C. Fantacci, and M. Beard, "Generalized labeled multi-bernoulli approximation of multi-object densities," *IEEE Transactions on Signal Processing*, vol. 63, no. 20, pp. 5487–5497, 2015.
- [29] H. Van Nguyen, H. Rezatofghi, B.-N. Vo, and D. C. Ranasinghe, "Online uav path planning for joint detection and tracking of multiple radio-tagged objects," *IEEE Transactions on Signal Processing*, vol. 67, no. 20, pp. 5365–5379, 2019.
- [30] C. Fantacci, B.-N. Vo, B.-T. Vo, G. Battistelli, and L. Chisci, "Robust fusion for multisensor multiobject tracking," *IEEE Signal Processing Letters*, vol. 25, no. 5, pp. 640–644, 2018.
- [31] H. Van Nguyen, H. Rezatofghi, B.-N. Vo, and D. C. Ranasinghe, "Distributed multi-object tracking under limited field of view sensors," *IEEE Transactions on Signal Processing*, vol. 69, pp. 5329–5344, 2021.

- [32] M. Beard, B.-T. Vo, B.-N. Vo, and S. Arulampalam, "Void probabilities and cauchy-schwarz divergence for generalized labeled multi-bernoulli models," *IEEE Transactions on Signal Processing*, vol. 65, no. 19, pp. 5047–5061, 2017.
- [33] H. Van Nguyen, B.-N. Vo, B.-T. Vo, H. Rezatofighi, and D. C. Ranasinghe, "Multi-objective multi-agent planning for discovering and tracking multiple mobile objects," *IEEE Transactions on Signal Processing*, 2024.
- [34] H. Van Nguyen, H. Rezatofighi, B.-N. Vo, and D. C. Ranasinghe, "Multi-objective multi-agent planning for jointly discovering and tracking mobile objects," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 05, 2020, pp. 7227–7235.
- [35] B.-N. Vo, B.-T. Vo, and M. Beard, "Multi-sensor multi-object tracking with the generalized labeled multi-bernoulli filter," *IEEE Transactions on Signal Processing*, vol. 67, no. 23, pp. 5952–5967, 2019.
- [36] M. Beard, B. T. Vo, and B.-N. Vo, "A solution for large-scale multi-object tracking," *IEEE Transactions on Signal Processing*, vol. 68, pp. 2754–2769, 2020.
- [37] A. Buonaviri, M. York, K. LeGrand, and J. Meub, "Survey of challenges in labeled random finite set distributed multi-sensor multi-object tracking," in *2019 IEEE Aerospace Conference*, 2019, pp. 1–12.
- [38] Y. Zhou, G. Battistelli, L. Chisci, L. Gao, G. Li, and P. Wei, "Median-based resilient multi-object fusion with application to lmb densities," *IEEE Transactions on Signal and Information Processing over Networks*, 2024.
- [39] L. Gao, G. Battistelli, and L. Chisci, "Resilient labeled multi-bernoulli fusion with peer-to-peer sensor network," *Information Fusion*, vol. 100, p. 101965, 2023.
- [40] M. Chamseddine, J. Rambach, D. Stricker, and O. Wasenmuller, "Ghost target detection in 3d radar data using point cloud based deep neural network," in *2020 25th International Conference on Pattern Recognition (ICPR)*, 2021, pp. 10 398–10 403.
- [41] B.-N. Vo and B.-T. Vo, "An implementation of the multi-sensor generalized labeled multi-bernoulli filter via gibbs sampling," in *2017 20th International Conference on Information Fusion (Fusion)*. IEEE, 2017, pp. 1–8.
- [42] N. Forti, G. Battistelli, L. Chisci, S. Li, B. Wang, and B. Sinopoli, "Distributed joint attack detection and secure state estimation," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 4, pp. 96–110, 3 2018.
- [43] Y. Nakahira and Y. Mo, "Attack-resilient $\mathcal{H}_2$, $\mathcal{H}_\infty$, and ℓ_1 state estimator," *IEEE Transactions on Automatic Control*, vol. 63, no. 12, pp. 4353–4360, 2018.
- [44] W. Yang, Y. Zhang, G. Chen, C. Yang, and L. Shi, "Distributed filtering under false data injection attacks," *Automatica*, vol. 102, pp. 34–44, 2019.
- [45] G. Ramos, D. Silvestre, and C. Silvestre, "General resilient consensus algorithms," *International Journal of Control*, vol. 95, no. 6, pp. 1482–1496, 2022.
- [46] Y. Chen, S. Kar, and J. M. Moura, "Resilient distributed estimation: Sensor attacks," *IEEE Transactions on Automatic Control*, vol. 64, no. 9, pp. 3772–3779, 2018.
- [47] J. Li, W. Abbas, and X. Koutsoukos, "Resilient distributed diffusion in networks with adversaries," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 6, pp. 1–17, 2019.
- [48] J. Huang, Y. Tang, W. Yang, and F. Li, "Resilient consensus-based distributed filtering: Convergence analysis under stealthy attacks," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 7, pp. 4878–4888, 2019.
- [49] M. Lei, H. Song, J. Fan, D. Xiao, D. Ai, Y. Gu, and J. Yang, "Gaa: Ghost adversarial attack for object tracking," *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2024.
- [50] V. Terzija, G. Valverde, D. Cai, P. Regulski, V. Madani, J. Fitch, S. Skok, M. M. Begovic, and A. Phadke, "Wide-area monitoring, protection, and control of future electric power networks," *Proceedings of the IEEE*, vol. 99, no. 1, pp. 80–93, 2010.
- [51] L. Xie, D.-H. Choi, S. Kar, and H. V. Poor, "Fully distributed state estimation for wide-area monitoring systems," *IEEE Transactions on Smart Grid*, vol. 3, no. 3, pp. 1154–1169, 2012.
- [52] S. Reuter, B.-T. Vo, B.-N. Vo, and K. Dietmayer, "The labeled multi-bernoulli filter," *IEEE Transactions on Signal Processing*, vol. 62, no. 12, pp. 3246–3260, 2014.
- [53] S. Reuter, A. Danzer, M. Stübler, A. Scheel, and K. Granström, "A fast implementation of the labeled multi-bernoulli filter using gibbs sampling," in *2017 IEEE Intelligent Vehicles Symposium (IV)*, 2017, pp. 765–772.
- [54] Y. Zhou, G. Battistelli, L. Chisci, L. Gao, G. Li, and P. Wei, "Median-based resilient multi-object fusion with application to lmb densities," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 10, pp. 473–486, 2024.
- [55] L. Gao, G. Battistelli, and L. Chisci, "Fusion of labeled rfs densities with different fields of view," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 6, pp. 5908–5924, 2022.
- [56] —, "Resilient labeled multi-bernoulli fusion with peer-to-peer sensor network," *Information Fusion*, vol. 100, p. 101965, 2023.
- [57] A. E. Abbas, "A kullback-leibler view of linear and log-linear pools," *Decision Analysis*, vol. 6, no. 1, pp. 25–37, 2009.
- [58] F. Nielsen, "Fast approximations of the jeffreys divergence between univariate gaussian mixtures via mixture conversions to exponential-polynomial distributions," *Entropy*, vol. 23, no. 11, p. 1417, 2021.
- [59] H. J. LeBlanc, H. Zhang, X. Koutsoukos, and S. Sundaram, "Resilient asymptotic consensus in robust networks," *IEEE Journal on Selected Areas in Communications*, vol. 31, no. 4, pp. 766–781, 2013.
- [60] A. Ghorbani, N. Nassir, P. Lavieri, and V. Ghorbani, "Binn: Behaviourally-informed neural network for discrete choice modelling," *Available at SSRN 5026729*.
- [61] A. Ghorbani, N. Nassir, P. S. Lavieri, P. B. Beeramoole, and A. Paz, "Enhanced utility estimation algorithm for discrete choice models in travel demand forecasting," *Transportation*, pp. 1–28, 2025.
- [62] M. Beard, B. T. Vo, and B.-N. Vo, "Ospa (2): Using the ospa metric to evaluate multi-target tracking performance," in *2017 International Conference on Control, Automation and Information Sciences (ICCAIS)*. IEEE, 2017, pp. 86–91.
- [63] D. Schuhmacher, B.-T. Vo, and B.-N. Vo, "A consistent metric for performance evaluation of multi-object filters," *IEEE transactions on signal processing*, vol. 56, no. 8, pp. 3447–3457, 2008.
- [64] J. Klupacs, A. K. Gostar, A. Bab-Hadiashar, and R. Hoseinnezhad, "Enhancing multi-object tracking through distributed information fusion in connected vehicle networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 15 897–15 908, 2024.